

Education

George Mason University, MS in Computer Science

May 2025

Fairfax, VA, USA

Relevant Coursework: Data Structures Algorithms, Secure Software Design, Machine Learning, Deep Learning, Object Oriented Design

Guru Gobind Singh Indraprastha University, B.Tech in Information Technology

Jul 2023

New Delhi, India

Experience

Software Developer, *College of Public Health, GMU*

June 2025 – Present

- Implemented a secure Flask-based app for **bruise image data set segregation** to assist department-wide research.
- Developed frontend in **React(Next.js)**, **REST API integrations**, and robust **secure** backend for managing bruise image data.
- Working on the backend of a **Bruise Detection Application** using YOLO; integrating 3D human body(Unity) visualization to display patient injuries for both web platform and **mobile application**.
- Application intended for use by healthcare professionals and legal proceedings related to domestic violence cases.

Software Developer & Research Assistant, *Geoweaver – NASA/ESIP*

Feb – Oct 2024

- Led 4 interns on ML workflows for geospatial data and co-authored their first publications.
- Added support for **PostgreSQL**, **MySQL**, and **H2** with CLI integration.
- Implemented an in-app code editor in **React** with **SpringBoot** as backend and improved UI responsiveness by 70%.
- Reduced processing time by 25% in Snow/Ozone Estimation models and ran large-scale workflows on Hopper supercomputer with 20% speed-up.
- Introduced **Vitest** for unit testing, accelerating development feedback loops and reducing CI runtime.

Software Developer, *Tech Solutions*

Aug 2022 – Jul 2023

- Led the development of **Android** features using **React Native**, integrating **real-time API** updates for inventory and order tracking.
- Refactored key components for performance and UI responsiveness, improving average **app load time by 25%**.
- Expanded **RESTful API** capabilities and implemented authentication middleware using JWT for secure user sessions.
- Designed and maintained **CI/CD pipelines** using GitHub Actions, automating testing, builds, and deployments across staging and production.
- **Mentored two interns** on React Native best practices, Git workflows, and backend integration, fostering a stronger development process.

Data Science Intern, *QBurst Technologies*

Jul – Aug 2022

- Assisted in Improving BigQuery query performance by 15% and automated migration from RedShift to GCP BigQuery using Python-based ETL.
- Authored internal documentation and training materials, boosting data engineering team throughput.

ML Intern, *DynPro*

Apr – Jun 2022

- Optimized recommendation pipeline (6.5% accuracy gain) through hyperparameter tuning and feature selection.
- Delivered regression and clustering results boosting success rate by 20%.

Software Engineer Intern, *Accops Systems*

Jan – Mar 2022

- Developed backend modules in Python and C++ for enterprise remote access system, reducing defect resolution time by 15%.

Cloud Trainee, *Tech Explica*

Jul – Aug 2021

- Deployed static sites on AWS (EC2, S3) and implemented CI/CD via GitHub Actions, improving deployment throughput by 20%.

Software Developer Intern, *Tech Solutions*

Jan 2020 – Jun 2020

- Assisted in developing RESTful APIs using Node.js and Express for a multi-tenant inventory management platform.
- Designed PostgreSQL database schemas and contributed to query optimization tasks under senior developer mentorship.

Publication

- Socioeconomic and Industrial Factors Influencing U.S. Ozone Pollution – NASA ESIP (AGU).
- Snow Water Equivalent Estimation Using ML & Remote Sensing – NASA ESIP.

Projects

Mason Critique – *2nd Place*

HackFAX 2025

- Developed a full-stack app using **React (Node.js, TypeScript)**, **Firebase**, and **Tailwind CSS**, enabling GMU students to review courses and professors with real-time feedback.
- Engineered back-end with **Firestore** and **Google OAuth**, integrating GPA calculators and **Chart.js** visualizations.

FinSight – *GenAI Finance Advisor (Google Capstone)*

Apr – May 2024

- Full-stack finance platform with **React**, **FastAPI**, and **Vertex AI Gemini**.
- Used document-grounded RAG and interactive **Recharts** for insights.

Skills and Tools

- **Languages:** Python, C++, Java, JavaScript, SQL, Bash
- **Frameworks/Tools:** Flask, FastAPI, React, React Native, Angular, TensorFlow, PyTorch,
- **Cloud:** AWS, Azure, GCP
- **Tools:** Docker, Kubernetes, GitHub Actions, Terraform, Linux